$$\sigma_n = \begin{pmatrix} n_z & (n_x - in_y) \\ (n_x + in_y) & -n_z \end{pmatrix}$$

SPIN OPERATOR MULTIPLICATION TABLES

A word about notation: Table 3 below uses the symbol *i* in two different ways. Inside a ket, such as |*io*, it is part of a state-label—*io* signifies "*in-out*." But when *i* appears outside of a ket symbol, as in *i*|*oo*>, it signifies the unit imaginary number.

Table 1: *Up-Down Basis*

	2-Spin Eigenvectors			
	$\|\mathbf{uu}\rangle$	$\|\mathbf{ud}\rangle$	$\|\mathbf{du}\rangle$	$\|\mathbf{dd}\rangle$
$\sigma_\mathbf{z}$	$\|uu\rangle$	$\|ud\rangle$	$-\|du\rangle$	$-\|dd\rangle$
$\sigma_\mathbf{x}$	$\|du\rangle$	$\|dd\rangle$	$\|uu\rangle$	$\|ud\rangle$
$\sigma_\mathbf{y}$	$i\|du\rangle$	$i\|dd\rangle$	$-i\|uu\rangle$	$-i\|ud\rangle$
$\tau_\mathbf{z}$	$\|uu\rangle$	$-\|ud\rangle$	$\|du\rangle$	$-\|dd\rangle$
$\tau_\mathbf{x}$	$\|ud\rangle$	$\|uu\rangle$	$\|dd\rangle$	$\|du\rangle$
$\tau_\mathbf{y}$	$i\|ud\rangle$	$-i\|uu\rangle$	$i\|dd\rangle$	$-i\|du\rangle$

Table 2: *Right-Left* Basis

	2-Spin Eigenvectors			
	$\|\mathbf{rr}\rangle$	$\|\mathbf{rl}\rangle$	$\|\mathbf{lr}\rangle$	$\|\mathbf{ll}\rangle$
$\sigma_\mathbf{z}$	$\|lr\rangle$	$\|ll\rangle$	$\|rr\rangle$	$\|rl\rangle$
$\sigma_\mathbf{x}$	$\|rr\rangle$	$\|rl\rangle$	$-\|lr\rangle$	$-\|ll\rangle$
$\sigma_\mathbf{y}$	$-i\|lr\rangle$	$-i\|ll\rangle$	$i\|rr\rangle$	$i\|rl\rangle$
$\tau_\mathbf{z}$	$\|rl\rangle$	$\|rr\rangle$	$\|ll\rangle$	$\|lr\rangle$
$\tau_\mathbf{x}$	$\|rr\rangle$	$-\|rl\rangle$	$\|lr\rangle$	$-\|ll\rangle$
$\tau_\mathbf{y}$	$-i\|rl\rangle$	$i\|rr\rangle$	$-i\|ll\rangle$	$i\|lr\rangle$